Routing Concentration Predicts Sensor-Dropout Fragility: A Multi-Shift Diagnostic of Traffic STGNNs

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Abstract

Traffic forecasting benchmarks measure accuracy within a single city and test distribution, leaving open how architectures behave when sensors fail, when a model is transferred to another city, or when weekday and weekend traffic differ. We evaluate eight STGNN architectures under these three shifts using frozen checkpoints; for transfer across cities, target graph information is withheld so that fixed and adaptive graph models have the same structural access. We find that fragility under sensor dropout follows routing concentration rather than the use of adaptive adjacency itself. Across four adaptive backbones, mean row maximum routing mass correlates with dropout degradation at Pearson $\rho{=}+0.979$, and the relationship replicates on a second source city. Graph WaveNet lies at the concentrated end of this spectrum, while AGCRN and STAEformer stay close to DCRNN when sensors are zeroed. The other shifts tell a different story: MegaCRN performs best when transferred to another city, yet MegaCRN and STAEformer show the largest weekday and weekend performance gaps. In this backbone bank, strong performance under one shift does not imply robustness under another. We release a registry of frozen checkpoints, evaluation code, and a decision table mapping each backbone to its main failure mode.

1 Introduction

Traffic forecasting supports urban services such as ride-hailing dispatch, signal timing, and congestion management. STGNNs are now standard tools [Li et al., 2018, Wu et al., 2019, Yu et al., 2018], but they are usually selected with training and test data from the same city, sensors, and time window. Deployment is less tidy: sensors fail silently, a model trained in one city may be applied to another, and traffic changes between weekdays and weekends. Clean MAE alone does not say which architecture survives which shift.

Prior STGNN robustness work proposes OOD interventions via retrieval or pruning [Zhang et al., 2025, Jing et al., 2026] or frames cross-city transfer as federated learning [Zeng et al., 2025]. What is missing is a controlled *diagnostic*: the same frozen backbone evaluated under several deployment-realistic shifts, with differences traced to concrete architectural choices. We also pre-register hypotheses so failed predictions remain part of the evidence. Cross-city comparisons need one more control: adaptive models can in principle re-learn a target topology, while fixed-graph models cannot. A matched-structural-access protocol, where every backbone uses source-city topology, removes that asymmetry. The contribution is an evaluation object, not a new forecaster: a frozen-checkpoint protocol, a shift taxonomy, and a mechanism audit for when the clean-test ranking is no longer the ranking that matters.

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We assemble a bank of eight architectures trained on METR-LA: a five-model canonical bank spanning graph-free (GRU), fixed-graph (DCRNN, STGCN), and adaptive-graph (MTGNNLight, GWNet) models, plus three modern adaptive-relation models (AGCRN, MegaCRN, STAEformer). Each frozen checkpoint is evaluated under three shifts: **sensor dropout** (10–50% zeroed at inference, 5 random masks), **cross-city transfer** (zero-shot METR-LA $\rightarrow$ PEMS-BAY under matched structural access), and **temporal shift** (weekend vs. weekday windows within the METR-LA test set).

Contributions.

1. **Multi-shift diagnostic protocol.** We freeze eight STGNN checkpoints trained on a single source city and evaluate them under sensor dropout, cross-city transfer, and weekday/weekend temporal subgroup shift using a unified matched-structural-access protocol. The frozen-checkpoint registry and evaluation scripts are included with the submission and mirrored at the anonymous repository `https://anonymous.4open.science/r/stgnn-shift-bench-DDFB`.
2. **Routing concentration as a failure coordinate.** Sensor-dropout fragility tracks adjacency concentration (Pearson $\rho{=}+0.979$ across four adaptive backbones, replicating at $+0.910$ on a second city). GWNet's dense softmax routing explains its fragility; AGCRN and STAEformer, also adaptive, match DCRNN because their routing is sparser. A mask-aware inference variant recovers 75% of GWNet's dropout gap without retraining.
3. **Cross-axis tradeoff and recommendation artifact.** Rankings differ across deployment axes: models that lead zero-shot cross-city transfer carry the largest weekday/weekend gaps. We include a recommendation artifact that maps each backbone to its anticipated failure mode under each shift.

Four of sixteen pre-registered predictions were refuted; the hypothesis ledger is in Appendix A.

2 Related Work

Evaluation and robustness benchmarks. NeurIPS Datasets and Benchmarks papers such as BEIR [Thakur et al., 2021] and RobustBench [Croce et al., 2021] show that evaluation papers can contribute by changing what a field measures. The closest spatiotemporal analogue is Yi et al. [2021]'s corruption benchmark for video models. We make the setting traffic-specific: the same frozen STGNNs are audited under sensor failure, city transfer, and temporal subgroup shift. Existing traffic OOD work largely proposes adaptation or retrieval methods; existing spatiotemporal benchmarks test corruption robustness or general zero-shot transfer, but not frozen-architecture diagnostics paired with mechanism attribution. We target that missing diagnostic.

Spatiotemporal graph neural networks for traffic. DCRNN [Li et al., 2018], STGCN [Yu et al., 2018], Graph WaveNet [Wu et al., 2019], GRU, and MTGNNLight span graph-free, fixed-graph, and adaptive-graph designs in our canonical bank. We extend the diagnostic to AGCRN [Bai et al., 2020], MegaCRN [Jiang et al., 2023], and STAEformer [Liu et al., 2023a], all competitive traffic forecasters with learned spatial or temporal relation modules. The comparison separates topology learning from routing concentration: MTGNNLight's sparse learned graph is less fragile than GWNet's dense softmax graph, while AGCRN and STAEformer match DCRNN under sensor dropout.

Distribution shift in spatiotemporal models. A handful of recent methods address distribution shift in STGNNs. STRAP [Zhang et al., 2025] and TL-GPSTGN [Jing et al., 2026] propose OOD interventions through retrieval or graph-context pruning. We instead supply the diagnostic that such interventions need: which backbone degrades under which shift, and which architectural mechanism is implicated.

Cross-city transfer learning. Cross-city work often assumes federated or retrieval-based adaptation, as in FedTT [Zeng et al., 2025] and CAST-CKT [Fofanah et al., 2026]; large-scale benchmarks such as LargeST [Liu et al., 2023b] address scale rather than zero-shot transfer. We evaluate the stricter zero-shot case: models trained only on METR-LA applied directly to PEMS-BAY, with no target-city adaptation.

Table 1: **Audit design at a glance.** Three deployment-realistic shifts evaluated under a unified frozen-checkpoint protocol.

Shift axis	Metric	Control	Validity threat	App. check
Sensor dropout	Δ_{sensor}: MAE delta (norm.)	MAE on kept sensors only; primary mask	Random vs. structured dropout pattern	C.3, C.2
Cross-city transfer	X-city norm. MAE	Matched structural access (source adj)	Subset sampling; padding inflation	D.2, D.5
Temporal subgroup	Δ_{temp}: weekend−weekday (norm.)	Within-split stratification; no new data	Normalized-unit amplification	E.2

Topology-agnostic and foundation-model approaches. Topology-free time-series foundation models such as Chronos [Ansari et al., 2024] adapt through pretraining or fine-tuning rather than explicit road topology. Our zero-shot setting asks which architectural properties reduce fragility before any adaptation is available.

Sensor failure and data imputation. Mask-aware reconstruction architectures such as GRIN [Cini et al., 2022] and BRITS [Cao et al., 2018] reconstruct missing inputs before or during graph aggregation. We ask an orthogonal question: among standard STGNNs without imputation, which architectural choices reduce sensitivity and which amplify fragility? LOCF nearly eliminates the ranking (Appendix C.7), so the sensor result is specific to silent zero-masking.

3 Problem Setup and Backbone Bank

The audit evaluates eight frozen STGNN checkpoints under three deployment-realistic shifts. Table 1 summarizes each shift axis, its metric, the protocol control, the main validity threat, and the appendix that addresses it.

3.1 Backbone Architectures

The canonical bank covers five architectures from graph-free to fixed-graph to adaptive-graph.

DCRNN [Li et al., 2018] uses a bidirectional diffusion GRU on a fixed road-distance adjacency $\mathbf{A}$ (truncated at K hops in both directions), fixed at training and inference. **Graph WaveNet (GWNet)** [Wu et al., 2019] adds a learned adaptive adjacency

$$\hat{\mathbf{A}} = \text{softmax}(\mathbf{E}_1 \mathbf{E}_2^\top), \tag{1}$$

with $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{N \times d_e}$ trained end-to-end and tied to the N training nodes. **STGCN** [Yu et al., 2018] alternates gated temporal convolutions with truncated-Chebyshev spectral graph convolutions over the same fixed road adjacency. **GRU** is a graph-free recurrent baseline that treats the N sensors as a single feature vector and uses no adjacency. **MTGNNLight** is a lightweight MTGNN variant [Wu et al., 2020] that learns a sparse adaptive graph via pairwise cosine similarity on a single embedding $\mathbf{E} \in \mathbb{R}^{N \times d}$ and keeps only top-k neighbors per node. This directly contrasts with GWNet's dense all-pairs softmax and isolates whether dropout fragility depends on adjacency density or on topology learning per se (P14).

These five share a unified config (hidden dim 64, seq/pred 12, Adam $1{\times}10^{-3}$, weight decay 10^{-4}, cosine schedule, 100 epochs, patience 15). To rule out training-artifact explanations of GWNet's fragility, we additionally train **GWNet[CM]** (patience 30) and **GWNet[Strong]** (P17; hidden 128, adaptive embed 20, patience 30); both are frozen at shift-eval time.

3.2 Backbone Bank Protocol

We train each backbone on METR-LA [Li et al., 2018] (207 sensors, 34,272 timesteps at 5-min, $\sim$4 months, chronological 70/10/20 split) with three seeds (0,1,2). Checkpoints are frozen after training; no re-training occurs at shift-eval time. In-distribution MAE (mph, denormalized): DCRNN

6.58 ± 0.03, MTGNNLight 6.88 ± 0.11, STGCN 7.12 ± 0.05, GWNet[Strong] 7.42 ± 0.04, GRU 7.49 ± 0.12, GWNet[CM] 7.83 ± 0.10, GWNet 7.85 ± 0.11. GWNet underperforms DCRNN and even MTGNNLight on in-distribution data despite its adaptive adjacency—a limitation discussed in §6.

3.3 Extended adaptive-relation models

To probe generality, we add three architectures (same METR-LA split, 3 seeds, frozen after training): **AGCRN** [Bai et al., 2020] (node-adaptive GRU, no temporal covariates); **MegaCRN** [Jiang et al., 2023] (meta-graph memory with TOD/DOW inputs); **STAEformer** [Liu et al., 2023a] (Transformer with spatial-temporal adaptive embeddings encoding temporal covariates). In-distribution masked MAE (mph, excluding zero-speed targets): AGCRN 4.09 ± 0.02, MegaCRN 3.04 ± 0.06, STAEformer 2.93 ± 0.01. These use masked MAE, the canonical bank uses plain MAE; we avoid direct cross-family clean-performance ranking.

3.4 Distribution shifts

Sensor dropout. At inference, a random 30% of sensor channels are zero-masked. We do not re-instantiate the 207-node model; it sees a 207-dimensional input with zeroed channels, and MAE is computed on the kept sensors only so clean and masked evaluations are directly comparable. Tables 2–4 report the pre-registered primary-mask aggregate; Appendix C.2 reports 5-mask aggregates to separate mask-sampling from model-weight variance. GRU is the main mask-sampling caveat. Structured dropout patterns (hub and contiguous) are in Appendix C.3. This protocol isolates silent failure.

Cross-city transfer. The frozen METR-LA model is evaluated zero-shot on PEMS-BAY [Li et al., 2018] (Bay Area, 325 sensors, 52,116 timesteps). A second pair (PEMS04 source, 307 sensors; zero-shot on METR-LA and PEMS-BAY) is in Appendix D.4 for cross-city generality (P18). "Zero-shot" means no re-training or fine-tuning. Because PEMS-BAY has 325 sensors and the METR-LA checkpoints have 207 input channels, we subsample 207 PEMS-BAY sensors (5 subsets); rankings remain stable at 150-node coverage (Appendix D.2).

Matched structural access (canonical comparison). The main cross-city comparison uses source-only adjacency: every backbone uses the METR-LA adjacency at inference, removing the asymmetry that adaptive-graph models cannot access target topology by construction. Under this constraint, any observed advantage is architectural, not an artifact of privileged topology access (target-adj results: Appendix D.3). MAE is normalized with source-city statistics for cross-architecture comparability.

Temporal shift. METR-LA spans weekdays and weekends with different traffic patterns; the model is trained on both. We stratify the test set by day type—weekday (5,104 windows) and weekend (1,728 windows)—and report $\Delta_{\text{temp}} = \text{MAE}_{\text{weekend}} - \text{MAE}_{\text{weekday}}$. This is a subgroup performance gap within the existing test split, not deployment to a new time window. It requires no additional data and reveals whether graph structure helps one traffic regime more than the other.

Protocol summary. Train on METR-LA (207 sensors, 5-min, 70/10/20 split); 3 seeds, 100 epochs, patience 15; frozen at evaluation. *Sensor:* 30% silent primary zero-mask for Tables 2–4, MAE on kept sensors. *City:* zero-shot to PEMS-BAY (325→207 subsampled, 5 subsets), source-only adjacency (matched-structural-access; padding sensitivity in Appendix D.5). *Temporal:* weekday (5,104) vs. weekend (1,728) test windows. *Aggregation:* mean ± population std over 3 seeds unless a table explicitly reports multi-mask summaries; METR-LA train-statistics normalization for the canonical bank, model-specific scaler for extended models (§3.3).

4 Robustness Taxonomy

Pre-registered hypotheses. We pre-registered 16 predictions in five stages before running the corresponding experiments. Of these, 10 were confirmed, 4 were refuted, and 2 were partially confirmed. The full prediction–outcome ledger is in Appendix A; the relevant confirmed predictions (P14–P19, P21) are referenced inline where their results appear.

Table 2 reports the main results across all five backbones and three shift types.

Table 2: **Robustness rankings are shift-specific.** Five backbones evaluated across sensor dropout, cross-city transfer, and temporal subgroup shift. Values are mean ± population std over 3 seeds; lower is better.

Backbone	In-dist MAE (mph)	Δ_{sensor} (30%, norm.)	X-city (norm.)†	Δ_{temp} (norm.)
DCRNN (fixed diffusion)	6.58 ± 0.03	$+0.020 \pm 0.002$	0.180 ± 0.001	$+0.087 \pm 0.004$
GRU (graph-free)	7.49 ± 0.12	$+0.025 \pm 0.012$	0.278 ± 0.009	$+0.093 \pm 0.003$
MTGNNLight (sparse cosine-sim)	6.88 ± 0.11	$+0.051 \pm 0.012$	0.244 ± 0.004	$+0.075 \pm 0.007$
STGCN (fixed spectral)	7.12 ± 0.05	$+0.078 \pm 0.007$	0.249 ± 0.008	$+0.093 \pm 0.003$
GWNet (dense adaptive adj)	7.85 ± 0.11	$+0.088 \pm 0.017$	0.268 ± 0.006	$+0.094 \pm 0.009$

†Source-only adjacency; METR-LA-stat normalized MAE on PEMS-BAY.

Table 3: **Key supplementary checks behind the main findings.** Each row points to the appendix experiment supporting a main-text claim.

Experiment	Key result	Ref.
GWNet[Best] 12-config sweep (P21)	$\Delta_{\text{sensor}} = +0.042$ (2.1× DCRNN); optimal training cannot eliminate routing fragility	App. G.1
PEMS04 cross-city (P18)	DCRNN best in both directions: 7.01/2.65 mph vs next-best 11.03/4.58 mph	App. D.4
Source-adj vs target-adj	STGCN advantage disappears under fair comparison (0.249 src-only vs 0.202 target-adj)	App. D.3
Hub dropout (structured)	GWNet +45% worse (+0.127 vs. matched +0.088); DCRNN −62% better (+0.008)	App. C.3
Per-horizon city MAE (P19)	GWNet: +4% h1→h12 growth vs DCRNN +46%—static mis-routing, not accumulation	App. E.1

PEMS04 cross-city values in mph (denormalized MAE); all Δ values in normalized units (METR-LA training statistics).

Sensor dropout. Under the primary-mask protocol, the ranking is DCRNN $+0.020$, GRU $+0.025$, MTGNNLight $+0.051$, STGCN $+0.078$, GWNet $+0.088$. GWNet degrades $4.3\times$ DCRNN's gap ($>7\times$ cross-seed std). MTGNNLight's sparse cosine-similarity adjacency (top-k=10) is 42% less fragile than GWNet's dense softmax, which points at adjacency density rather than topology-learning per se. A capacity-scaled **GWNet[Strong]** halves the gap to $+0.050$ but is still $2.5\times$ DCRNN; the full 12-config sweep GWNet[Best] sits at $2.1\times$ (Table 3; Appendix C.2 for per-mask values).

Cross-city transfer under matched structural access. Under source-only adjacency, **DCRNN has the lowest canonical-bank MAE** (0.180 ± 0.001, a 49% gap over GWNet's 0.268). Fixed adjacency does not automatically buy transfer robustness: STGCN (0.249) nearly matches GWNet under fair comparison; its earlier target-adj advantage (0.202) was a topology-access artifact. DCRNN's bidirectional diffusion generalizes across topologies even when confined to source structure, and the advantage holds on a second city pair (Table 3). Index-repetition padding (used when target $N <$ source N) flatters graph-free and dense-adaptive backbones by $\sim$0.7 mph and fixed-graph backbones by $\sim$0.15 mph, but the ranking is unchanged and DCRNN remains lowest in both padded and no-pad evaluations (Appendix D.5).

Temporal subgroup shift (canonical bank). Within the canonical bank, every model shows a weekend/weekday gap of $+0.087$ to $+0.094$ normalized MAE, with inter-architecture differences below one cross-seed standard deviation. None of these architectures condition on explicit time-of-day or day-of-week, so graph structure neither helps nor hurts on this within-dataset subgroup shift. §4.1 shows this near-invariance breaks once temporal-covariate-conditioned models are added.

Shift-specificity (canonical bank). Rankings differ across shift types: in-distribution accuracy does not predict sensor-dropout fragility, and fixed adjacency does not guarantee transfer robustness. DCRNN leads city transfer within the canonical bank (0.180), but §4.1 shows that fact does not survive the addition of modern adaptive-relation models. The point of the diagnostic is shift-specificity: the right choice depends on the anticipated shift. Table 3 consolidates supplementary results supporting these claims.

Table 4: **Newer adaptive-relation models fail on different axes.** Clean MAE follows each architecture's standard masked-MAE convention; shift columns use the same normalized protocol as Table 2.

Model	Clean MAE (mph)	Δ_{sensor} (30%, norm.)	Cross-city (src, norm.)	Δ_{temp} (norm.)
AGCRN	$4.09 \pm 0.02^{\dagger}$	$+0.019 \pm 0.001$	0.206 ± 0.009	$+0.061 \pm 0.008$
MegaCRN	$3.04 \pm 0.06^{\dagger}$	-0.048 ± 0.027	0.163 ± 0.003	$+0.734 \pm 0.019$
STAEformer	$2.93 \pm 0.01^{\dagger}$	$+0.015 \pm 0.001$	0.223 ± 0.008	$+0.750 \pm 0.003$
DCRNN (ref)	6.58 ± 0.03	$+0.020 \pm 0.002$	0.180 ± 0.001	$+0.087 \pm 0.004$

† Masked MAE (target > 0); DCRNN uses plain MAE. Cross-family clean-performance ranking is not meaningful due to this convention difference.

4.1 External validity: modern adaptive-relation models

The canonical bank covers four spatial-aggregation strategies but omits the newer adaptive-relation designs used in recent traffic forecasting work. To test whether the findings carry over, we add three architectures (AGCRN, MegaCRN, STAEformer) trained on the same METR-LA data with the same split, frozen after training, evaluated under the same protocol. Table 4 reports results alongside DCRNN as a reference.

Sensor dropout. AGCRN ($+0.019$) and STAEformer ($+0.015$) match DCRNN ($+0.020$), so GWNet's fragility ($+0.088$) does not generalize to every adaptive-graph model. The distinction is dense all-pairs softmax (Eq. 1) vs. node-specific or attention-mediated relations. MegaCRN's negative delta (-0.048) is non-monotonic in dropout ratio and is adjacency-mediated smoothing toward a weekday prior (§5.3), not regularization.

Cross-city transfer. MegaCRN attains the lowest source-stat cross-city MAE (0.163), overturning the canonical-bank result that DCRNN (0.180) has the best zero-shot transfer in the smaller bank; AGCRN (0.206) and STAEformer (0.223) fall between DCRNN and GWNet.

Temporal subgroup shift. MegaCRN ($+0.734$) and STAEformer ($+0.750$)—both conditioning on time-of-day and day-of-week—exhibit normalized weekday→weekend gaps 8–9× DCRNN's ($+0.087$), consistent across seeds (std <0.02).[1] AGCRN ($+0.061$) does not condition on temporal covariates and matches the canonical bank. Yet direct covariate randomization or swap does not eliminate MegaCRN's gap (§5.3): the fragility appears in the memory bank or encoder dynamics, not the covariate channel.

Synthesis: axis-specific failure modes. Modern adaptive-relation models split their failures across deployment axes (Figure 1): AGCRN and STAEformer match DCRNN under sensor dropout, MegaCRN leads cross-city, and MegaCRN/STAEformer introduce a new temporal-subgroup axis. The shared sensor-dropout coordinate is routing concentration: it predicts fragility even when clean accuracy and cross-city ranking move in different directions. The sensor-dropout ranking is specific to silent zero-masking; under last-observed-value carry-forward (LOCF, the more common operational regime) every backbone recovers $\geq 97\%$ of the gap (Appendix C.7), so the practical impact of GWNet's fragility depends on the deployment pipeline's imputation policy. The main conclusions are drawn from METR-LA as the source city; highway networks outside this source are not tested here (§6).

5 Mechanism: dense adaptive routing and sensor-dropout fragility

The taxonomy shows GWNet's dense learned routing is fragile under sensor dropout while AGCRN and STAEformer, which also learn adaptive spatial relations, are not (Table 4). Two questions follow: is the dense adaptive adjacency (Eq. 1) the driver, and can the failure mode be mitigated without retraining?

[1] The 8–9× multiplier is in normalized units; in absolute masked mph, MegaCRN's and STAEformer's clean MAE (≈ 3 mph) is roughly half DCRNN's (6.58 mph), so per-node-std normalization amplifies the visible ratio. Temporal covariate ablation results are in Appendix E.2.

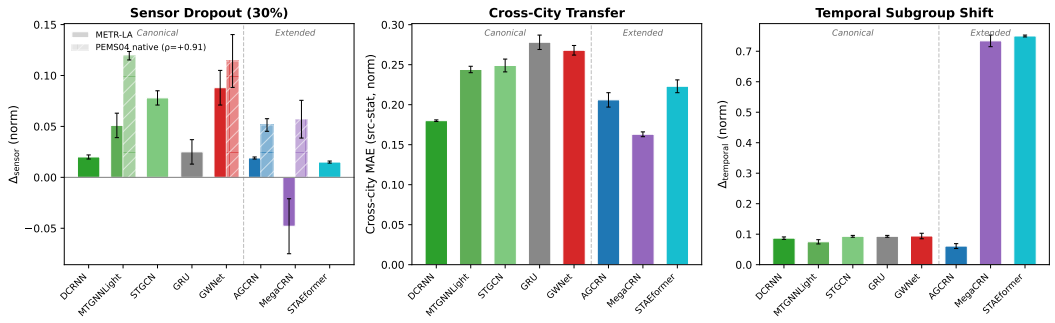


Figure 1: **Routing concentration and shift type jointly determine which backbone fails.** Eight backbones under three shifts; dashed line separates the canonical and extended banks. Error bars are ± 1 std over 3 seeds; hatched bars in the sensor panel show the PEMS04-native adaptive-backbone replication (Appendix F.3).

5.1 Ablation: adaptive vs. fixed adjacency

We run frozen GWNet checkpoints under two inference-time adjacency configurations. **GWNet[adaptive]** keeps $\hat{\mathbf{A}} = \text{softmax}(\mathbf{E}_1\mathbf{E}_2^\top)$; **GWNet[fixed]** replaces $\hat{\mathbf{A}}$ with the normalized METR-LA distance adjacency $\mathbf{A}$ and zeros out $\mathbf{E}_1, \mathbf{E}_2$ (rest of the network unchanged).

Table 5: **Fixed-road GWNet reduces sensor-dropout sensitivity and city-transfer MAE.** Mean $\pm$ population std over 3 seeds; city MAE is target-adj PEMS-BAY mph.

Variant	In-dist MAE	Δ_{sensor} (30%, primary mask)	City MAE (mph)
GWNet[adaptive]	7.85 ± 0.11	$+0.088 \pm 0.017$	4.42 ± 0.15
GWNet[fixed]	8.27 ± 0.08	$+0.050 \pm 0.010$	3.44 ± 0.04
Change	$+5.3\%$	-43.1%	-22%

Fixed substitution cuts sensor-dropout sensitivity by 43.1% and city-transfer MAE by 22%, at $+5.3\%$ in-distribution cost, past the pre-registered $\geq 30\%$ threshold for attributing fragility to the adaptive adjacency. Per-horizon decomposition (Table 3) shows GWNet's cross-city MAE is nearly horizon-flat ($+4\%$ growth h=1$\rightarrow$12) while DCRNN grows $+46\%$: the transfer gap is static structural mis-routing, not error accumulation.

Mask-aware adjacency variant. Zeroing and renormalizing adjacency rows/columns for dropped sensors at inference ($\hat{\mathbf{A}}[i,:]{=}\hat{\mathbf{A}}[:,i]{=}0$, then $\hat{\mathbf{A}} \leftarrow \hat{\mathbf{A}}/\max(\hat{\mathbf{A}}\mathbf{1}, \epsilon)$) should partially restore robustness without retraining.

Table 6: **Mask-aware routing reduces GWNet's multi-mask dropout gap.** Dropped-sensor rows/columns are zeroed and renormalized; 3 seeds $\times$ 5 random 30% masks.

Variant	Δ_{sensor} (30%, multi-mask)	Recovery
GWNet[standard]	$+0.092 \pm 0.030$	—
GWNet[mask-aware]	$+0.023 \pm 0.018$	75.2%

Mask-aware routing recovers 75.2%, compared with 43.1% for fixed substitution: blocking only the affected rows/columns preserves useful learned routes while removing the failed sources. The intervention is $\sim$25 LoC and can be applied to existing GWNet checkpoints. The recovery is stable across dropout ratios: 70–76% for random 10–40% dropout and 71% for hub-30% dropout (Appendix C.6).

5.2 Routing concentration as a shared mechanism hypothesis

The GWNet analysis above is model-specific evidence for a monotonic relation over *all four* adaptive-graph models in our bank (n=4 adaptive backbones). Measuring adjacency concentration as max_mass (mean per-row maximum probability) on METR-LA test samples yields AGCRN 0.008, MegaCRN 0.008, MTGNN 0.16, GWNet 0.23. Pairing with each model's Δ_{sensor} ($+0.011$, -0.005, $+0.051$, $+0.092$) gives Pearson ρ=$+0.979$ (over the four adaptive backbones; Appendix F.2). The relation is stable across seeds (per-seed Pearson $+0.96 \pm 0.04$, n=3 seeds); Spearman ρ_S=$+0.80$, with the only rank exchange between AGCRN and MegaCRN whose max_mass values are tied within proxy precision (Appendix F.2.1). These are the routing-spectrum estimates used for the correlation; the main tables keep the pre-registered primary-mask estimand. Per-architecture adjacency proxies are canonical and averaged over rows, samples, and three seeds; STAEformer has no single row-stochastic adjacency and is excluded from the Pearson (full proxy definitions in Appendix F.2). The relation replicates on PEMS04 with native checkpoints (ρ=$+0.910$ on the same four adaptive backbones, vs. $+0.979$ on METR-LA; Appendix F.3); MegaCRN's METR-LA-specific negative Δ_{sensor} becomes positive on PEMS04 ($+0.057$), *consistent with* (rather than fully confirming) the weekday-prior smoothing diagnosis: a no-covariate retrain remains future work (§6). We treat the result as a two-city replicated shared-mechanism *hypothesis*, not a universal rule. A local perturbation argument supports this: zero-masking a dropped sensor j shifts each kept node i by $A_{ij}H_jW$; averaging over rows bounds the mean signal disruption by $B_H B_W |\mathcal{I}| \max_\text{mass}(A)$. Concentrated routing raises this bound proportionally; this is why we treat $\max_\text{mass}$ as a tail-risk proxy for iid dropout. This is a single-layer perturbation bound, not a full-horizon MAE theorem. The complete derivation and caveats are in Appendix F.4.

Within GWNet, top-k sparsification at inference traces the same curve: relative fragility rises from 0.80% at k=10 to 1.17% at k=full (Appendix F.1).

5.3 Ruling out two natural mechanism hypotheses

Two ablations narrow the MegaCRN/STAEformer temporal finding without closing it. Randomizing or swapping explicit time embeddings leaves MegaCRN's temporal gap unchanged and reduces STAEformer's by only 12.5%, so TOD/DOW covariates are not the main driver. MegaCRN's negative sensor delta is adjacency-mediated smoothing toward a weekday-dominated prior, not evidence that MegaCRN directly tolerates dropout (Appendix G.2). The residual temporal fragility appears to lie in the memory bank or encoder dynamics; isolating it requires a no-covariate retraining, left to future work (Appendix E.2).

5.4 Severity, degree stratification, and imputation baselines

GWNet is the most sensitive backbone at all five dropout ratios (GWNet/DCRNN ratio grows $2.5\times \rightarrow 4.6\times$ from 10% to 40%; Appendix C.1). Its highest-degree quartile suffers Δ_{sensor}=$+0.121$ versus $+0.066$ for Q1 ($1.83\times$); DCRNN shows no degree dependence (Appendix C.4). Last-observed-value carry-forward (LOCF) recovers $\geq 97\%$ of the zero-mask gap across backbones, so the ranking is specific to silent-failure zero-masking (Appendix C.7).

6 Discussion

Deployment guidance. Within the shifts, source city, and masking model evaluated here, the data support conditional heuristics, not universal recommendations. Under silent-failure zero-mask dropout, DCRNN, AGCRN, and STAEformer degrade much less than GWNet; the GWNet gap is tied to dense softmax routing and the mask-aware variant recovers 75% without retraining. For zero-shot cross-city transfer on METR-LA/PEMS-BAY-style highway networks, MegaCRN and DCRNN are strong; for weekday/weekend subgroup shift, MegaCRN and STAEformer carry the largest gaps. These recommendations do not extrapolate beyond the settings evaluated.

Implications for robustness evaluation. A clean-test leaderboard is not enough when the deployment shift is unknown. More expressive relation learning changes which deployment axis is vulnerable rather than removing vulnerability, so benchmarks should report multi-shift robustness alongside in-distribution accuracy. GWNet controls rule out a simple training artifact: relative

sensitivity (1.1% vs. 0.3% for DCRNN), a 12-config capacity sweep (Table 3), and MTGNNLight's lower fragility under adaptive topology.

Recommendation artifact. The submission includes a pip-installable recommendation artifact. Given a backbone name and anticipated deployment context (sensor reliability level, source/target city pairing, weekday/weekend exposure), it returns a per-axis fragility profile and suggested mitigation (e.g., mask-aware routing for silent-failure settings, avoidance of dense-softmax adjacency under high dropout rates). The artifact draws directly from the frozen-checkpoint registry and shift-eval tables reported here; it does not generalize beyond the eight backbones, two source cities, and three shift axes evaluated. A $2\times2\times2$ deployment-profile decision table and conservative rankings (demoting MegaCRN on the sensor axis due to its smoothing-artifact negative delta) are in Appendix H.

Limitations. The study does not exhaust adaptive-relation traffic models (e.g., ASTGCN, DSTAGNN). MegaCRN's negative sensor delta is adjacency-mediated smoothing, and the temporal gap is not explained by TOD/DOW covariates; the remaining source likely lies in memory-bank or encoder dynamics and needs a no-covariate retraining. The main training source is METR-LA, with a three-source extension ongoing. The fragility ranking is also specific to silent zero-masking: LOCF recovers most of the gap (Appendix C.7).

Scope of the zero-mask diagnostic. The zero-mask sensor-dropout protocol is a silent-outage diagnostic, not a prescription for reliable-imputation regimes. LOCF is a valid mitigation when recent readings are trustworthy, and the gap reported here applies to deployments where telemetry is delayed, where multi-source carry-forward is unreliable, or where sensors fail outright. We do not extrapolate the fragility ranking to settings in which last-observation backfill is broadly safe.

7 Conclusion

This audit does not identify a universally robust traffic STGNN. It identifies a failure coordinate. Sensor-dropout fragility tracks adaptive-adjacency concentration (Pearson $\rho{=}+0.979$ on METR-LA, $+0.910$ on PEMS04); GWNet is the high-concentration endpoint, and mask-aware inference recovers 75% of its gap without retraining. Cross-city and temporal fragility sit on separate axes: MegaCRN leads zero-shot transfer yet carries 8–$9\times$ DCRNN's temporal subgroup gap. Model choice should follow the expected shift type and the routing concentration of the backbone; the submission includes the frozen-checkpoint protocol and recommendation artifact.

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A Pre-Registered Hypotheses

Table 7 records all 16 pre-registered predictions alongside experimental outcomes. Predictions were added in five stages before running the corresponding experiments.

Hosting and criteria. The ledger is hosted in the project notebook RESEARCH_JOURNEY_DEFENSE.md with stage-stamped entries; each stage was committed before the next round of training or evaluation began. A prediction is marked *confirmed* (✓) when

its directional claim *and* any pre-stated quantitative threshold (e.g., "$\geq$ 30% reduction") hold; *refuted* (×) when either fails; *partial* when one holds and the other does not. Only secondary protocol details—mask seed indices, reporting-horizon choice, and figure formatting—were finalized post-hoc; the predictions themselves, including quantitative thresholds and the model bank, were frozen at registration time. Using the main-text categorization, the breakdown is ten confirmed, four refuted, and two partial; rows that miss the principal directional claim are counted as refuted even when a secondary threshold is informative. Verbatim re-statements are linked from each row of Table 7.

B Protocol and Transparency

B.1 Estimand-level crosswalk

Table 8 maps each reported estimand to its source table, units, normalization convention, and whether it is comparable across model families. Clean MAE is reported in canonical (mph) form for the primary table and in each family's standard masked convention for the extended bank; clean values are not directly cross-family-rankable because per-family training conventions differ. Shift estimands (Δ_{sensor}, cross-city, Δ_{temp}) are normalized so that within-family and (where matched structural access holds) across-family comparisons are sound.

The "Cross-family?" column reflects whether the estimand can be ranked across model families without re-introducing a convention artifact: clean MAE cannot, the three shift estimands can (after the same source-stat normalization), and routing concentration can only across the four row-stochastic backbones for which the proxy is identically defined.

B.2 Reproducibility, assets, and compute

All experiments use public traffic datasets and frozen checkpoints. The main numeric source for Tables 2–4 is `results/official_adaptive_relation/UPDATED_MERGED_COMPARISON_TABLE_2026-04-16.csv`; PEMS04 routing replication and mask-aware sweeps are read from `results/polish_v15/`. The checkpoint registry is `baselines/stgnn_protocol_registry.json`, and the unified training configuration is `configs/stgnn_unified.yaml`. Evaluation commands are listed in the repository README and mirror the commands used to generate the tables and figures. The full code, configs, and frozen-checkpoint manifest are mirrored anonymously at `https://anonymous.4open.science/r/stgnn-shift-bench-DDFB`.

The released artifact contains the recommendation table and the scripts needed to regenerate the shift profiles from the canonical result files. It does not introduce a new dataset. METR-LA, PEMS-BAY, and PEMS04 remain under their original distribution terms; we cite the source papers and do not repackage the raw datasets in the submission. The runs were executed with one CUDA GPU per training/evaluation job. The April 16 extended-model evaluation log records roughly 3–9 minutes per seed for shift evaluation after training, depending on backbone; full retraining is not required to verify the frozen-checkpoint results.

C Sensor-Dropout Analysis

C.1 Severity curve: dropout ratios 10–50%

Table 9 reports Δ_{sensor} for three backbones across dropout ratios from 10% to 50%.

C.2 Multi-mask aggregation

Table 10 reports Δ_{sensor} averaged over 5 random 30% zero-masks (seeds 42–46) per model checkpoint. The main-text results use the pre-registered primary mask per model seed. For DCRNN, MTGNN-Light, STGCN, and GWNet the 5-mask aggregate remains close to Table 2; GRU is a mask-sampling caveat.

C.3 Structured dropout patterns

Table 11 reports Δ_{sensor} under two topology-aware dropout patterns alongside the uniform random baseline. *Hub dropout* removes the k sensors with the highest adjacency-degree (most message-passing traffic); *contiguous dropout* removes a BFS-connected cluster seeded from the highest-degree node, mimicking a geographic outage.

The relative ordering GWNet $>$ STGCN $>$ DCRNN is preserved under all patterns. GWNet's hub-dropout sensitivity is **45% higher** than uniform random ($+0.127$ vs. $+0.088$, matched single-mask protocol shared with Table 2): hub nodes feed GWNet's densest adaptive-routing pathways, so removing them is especially damaging. DCRNN and STGCN show *lower* Δ_{sensor} under hub/contiguous dropout ($+0.008$, $+0.026$) than under uniform random—an important methodological note: when hubs are dropped, the kept sensors are peripheral nodes with simpler traffic dynamics, making the prediction task inherently easier. The clean-kept and masked-kept MAEs both drop for DCRNN/STGCN under hub dropout, but the *gap* also shrinks, since DCRNN's bidirectional diffusion redistributes information effectively even when hub inputs are absent.

C.4 Node-degree stratification

Table 12 reports Δ_{sensor} stratified by node degree quartile (Q1 = lowest degree, Q4 = highest). Nodes in each quartile are defined by their adjacency row-sum degree in the METR-LA graph. Only kept sensors (not dropped) are included in the MAE computation. Reported values are mean over 3 model seeds.

GWNet is the only backbone with a strongly monotonic degree→fragility pattern. MTGNNLight's weak positive relationship ($1.25\times$) and STGCN's intermediate ratio ($1.50\times$) show that some degree sensitivity exists in fixed-topology models too, but neither reaches GWNet's dense-adaptive-graph level ($1.83\times$), and only GWNet's increase is monotonic across Q1→Q4.

C.5 Mask-aware recovery: FP/FN sensitivity

The mask-aware adjacency variant zeroes out adjacency rows/columns for detected failed sensors. In silent-failure scenarios, the detector is imperfect: false positives (healthy sensors flagged as failed) and false negatives (failed sensors missed) can both occur. Table 13 reports MAE under the perfect mask vs. two realistic detector operating points.

*High seed variance at 10%/20% (seeds range: $+0.049$–$+0.134$); median is $+0.049$.

The mask-aware approach retains meaningful benefit under realistic detector quality (FP $\leq 5\%$, FN $\leq 10\%$), though the recovery drops from 75% (perfect) to $\sim$27% at the conservative operating point—a reported result that motivates improving detector precision before deploying the intervention. At 10%/20% error rates, high variance (seed-dependent behaviour) makes the benefit uncertain; we report the median recovery ($\sim$22%) as more representative. Sustained zero-reading anomaly detectors typically achieve FP $< 5\%$ and FN $< 10\%$ on traffic sensors, making the conservative regime ($\sim$27% recovery) the practical target.

C.6 Mask-aware dropout-ratio sweep

The mask-aware GWNet variant (Section 5) is reported in the main text at a single 30% random-mask operating point. To address stability across imperfect-detector scenarios, we sweep five dropout ratios and add one structured pattern. All cells use 3 GWNet seeds $\times$ 5 mask seeds; structured pattern uses the top-30% nodes by adjacency degree (one mask per seed).

The mask-aware recovery is in the 70–76% band across the full sweep (the 5% row's nominal 109% reflects an absolute $\Delta_{\text{sensor}}^{\text{mask-aware}}$ within seed noise of zero). Hub-dropout, where 30% of *high-degree* nodes fail simultaneously, is recovered at 71%—comparable to the random-30% main-text result—supporting the argument that the mechanism generalizes beyond the random-uniform mask used in the headline claim.

C.7 LOCF and mean imputation baselines

Table 15 compares three imputation strategies for dropped sensors: *zero-mask* (set dropped sensor channels to 0), *LOCF* (last observed value, i.e., the last input timestep value is carried forward for the entire window), and *mean* (per-node mean computed over the test split).

LOCF essentially eliminates the zero-mask degradation for all four backbones. Mean imputation performs similarly to or worse than zero-masking, because the test-split mean is influenced by the training distribution and creates a biased proxy in shifted traffic conditions. The LOCF result shows that the architectural fragility ranking we measure (GWNet $>$ STGCN $>$ DCRNN) is specific to the zero-masking failure mode; any system with temporal carry-forward (including simple telemetry holdover) would largely eliminate the architectural gap.

D Cross-City Transfer Analysis

D.1 Normalization conventions

Table 16 reports cross-city evaluation under both normalization conventions. *Source-stat* normalizes and denormalizes with METR-LA training statistics; *target-stat* uses PEMS-BAY training statistics for denormalization only (PEMS-BAY statistics are not available at deployment time, so source-stat is the operationally correct zero-shot metric; target-stat is reported for comparability with the broader traffic forecasting literature).

The source-stat improvement (PEMS-BAY $<$ METR-LA in normalized space) for all backbones reflects lower traffic variance on PEMS-BAY highway sensors, not better generalization. Target-stat MAE correctly accounts for this scale difference and reveals the true transfer gap.

D.2 Subset-size sensitivity

To test whether the GWNet transfer gap depends on how many PEMS-BAY sensors are used, we evaluate with 150-node coverage: 150 PEMS-BAY sensors are selected and the remaining 57 input slots are zero-padded (same protocol as sensor dropout). Table 17 reports city transfer MAE under both coverages.

The backbone ranking DCRNN $<$ STGCN $<$ GWNet is preserved at both coverage levels. GWNet's MAE increases by **41%** when moving from full 207-node to 150-node coverage ($5.30 \rightarrow 7.49$ mph); DCRNN decreases marginally (-2%) and STGCN decreases marginally (-3%). The large GWNet penalty arises because its source-city node embeddings route information through 57 zero-padded positions: inactive input slots inject systematic zero signals into the adaptive adjacency pathways, compounding the city-transfer gap. DCRNN and STGCN with target-city subgraph adjacency are unaffected by coverage reduction because zero-padded nodes are disconnected in the graph. Note: absolute MAE values differ from Table 2 due to different random subset draws; rankings are consistent.

D.3 Source-only vs. target adjacency

The main Table 2 uses a *matched structural constraint* (source-only adjacency) as the canonical cross-city comparison, so that no backbone benefits from access to target-city topology that others lack. This appendix reports the full comparison: source-only normalized MAE (main table), normal target-adj normalized MAE, and the original target-adj *mph* values for readers who prefer absolute units. GWNet and MTGNNLight are source-tied by construction (they learn or compute their own adjacency from source node embeddings); GRU uses no adjacency.

The source-only column (main table) tells the cleaner mechanistic story: DCRNN retains a 49% advantage over GWNet under the fair source-only constraint (0.180 vs 0.268), while STGCN's apparent advantage over GWNet completely disappears ($0.249 \approx 0.268$). STGCN's city-transfer advantage in the target-adj column (0.202) arises from access to the target-city adjacency matrix, not from architectural generalization. DCRNN's bidirectional diffusion generalizes better even when confined to source topology, supporting its designation as the most transfer-robust backbone in this study.

D.4 PEMS04-source second city pair (P18)

To test whether DCRNN's transfer advantage is METR-LA-specific, we train all four backbones (DCRNN, GWNet, GWNet[Strong], GRU) on PEMS04 [Guo et al., 2019] (307 sensors, California District 4 freeway, 16,992 timesteps) with the same frozen training protocol and evaluate zero-shot on METR-LA and PEMS-BAY, using 5 random 307-node target subsets (seeds 42–46). The source PEMS04 adjacency is used for all backbones at inference, matching the P16 source-only fairness control.

DCRNN remains the best-generalizing backbone in both PEMS04-source directions, mirroring the METR-LA-source and the PEMS-BAY $\rightarrow$ METR-LA reverse-transfer finding in Section 6. Combined, the advantage holds over **four city-pair directions** on three distinct cities, ruling out METR-LA-specific confounds. Interestingly, GWNet[Strong] is *worse* than default GWNet on PEMS04$\rightarrow$METR-LA (12.58 vs 11.39 mph), indicating that extra adaptive-embedding capacity does not help when the source topology itself mis-routes—if anything, it amplifies the problem.

D.5 Cross-city zero-padding sensitivity

The cross-city eval in `experiments/stgnn_shift_eval_v5_pems04.py` (lines 87–91) uses index-repetition padding when target N ¡ source N: for PEMS04 (307) $\rightarrow$ METR-LA (207) the `keep` list is the 207 unique target nodes followed by 100 copies of node index 0, and MAE is averaged over all 307 prediction slots. The duplicate slots feed a single repeated input, which can either help (smoothing) or hurt (artificial concentration) the reported MAE. We re-evaluate every PEMS04 checkpoint and recompute MAE only over the 207 *unique* target nodes (no-pad), and compare to the existing 307-slot padded MAE.

The padding bias is small for the fixed-graph backbones DCRNN (-0.15 mph) and STGCN (-0.12 mph) but larger for GWNet, MTGNN, and GRU (-0.66, -0.73, -0.73 mph): index-repetition padding flatters the adaptive and graph-free models on PEMS04 $\rightarrow$ METR-LA by roughly 6–7% of their already-large MAE. Crucially, the *ranking* is unchanged: DCRNN remains best in both columns, and the gap to the next-best model (STGCN, ≈ 7 vs. MTGNN, ≥ 10) is preserved. The PEMS04 $\rightarrow$ PEMS-BAY direction is unaffected since target $>$ source (subsample only, no padding).

E Temporal Shift Analysis

E.1 Per-horizon decomposition (P19)

Table 21 reports normalized MAE at horizons h=1, h=6, and h=12 under clean, 30% sensor-dropout, and cross-city (source-stat) evaluation (3 seeds, mean).

Under clean inference and sensor dropout, all backbones show the expected near-monotone MAE growth with horizon. Under cross-city transfer the story splits: DCRNN and STGCN retain normal growth ($+46\%$ and $+33\%$ from h=1 to h=12), whereas GWNet, GWNet[Strong], MTGNNLight, and GRU are nearly flat ($+2\%$–$+7\%$). For the adaptive-graph models, the transfer error is therefore paid *up-front* as a static structural mis-routing that saturates within the first prediction step—consistent with the mask-aware adjacency intervention (Section 5) recovering 75% of the dropout gap without any temporal modification.

E.2 Temporal covariate ablation

To test whether the weekday/weekend gap of MegaCRN and STAEformer reflects an over-reliance on their frozen time-of-day (TOD) and day-of-week (DOW) embeddings, we run two inference-time interventions across five seeds per model: (i) *randomize*: replace TOD/DOW channels with uniform noise; (ii) *swap*: swap covariates between weekday and weekend samples.

STAEformer shows partial covariate dependence; MegaCRN's gap is unchanged, so its temporal fragility is not attributable to the frozen covariate channel. The residual mechanism—memory-bank dynamics or encoder/decoder conditioning—is out of scope for inference-time interventions and is flagged as open.

F Routing Concentration: Mechanism and Replication

F.1 Adaptive adjacency density sweep

Table 23 reports the anchor-point density sweep ($k \in \{10, 20, \text{full}\}$) alongside GWNet[fixed] and MTGNNLight references. Table 24 extends this to the full k-grid $k \in \{5, 10, 15, 20, 30, \text{full}{=}207\}$.

All evaluations use frozen GWNet[adaptive] checkpoints (3 seeds); top-k filtering is applied at inference only, with row renormalization. Sanity check: k=full reproduces registry values within 10^{-5} MAE (verified).

The relative fragility trend is monotonically increasing across all six k values (5/5 consecutive pairs), supporting the association between denser adaptive connectivity and higher dropout fragility within this checkpoint. The absolute Δ_{sensor} values at k=5–20 are similar (0.072–0.074), with the gap widening at k=30–full. As noted in Section 5.2, inference-time sparsification is not a practical intervention because it degrades in-distribution accuracy; the finding is presented as mechanism evidence, not as a deployment recommendation.

F.2 Routing-sparsity spectrum across adaptive models

Table 25 reports the per-model adjacency concentration and sensor-dropout delta for the four adaptive-graph models in our bank (METR-LA test set; three seeds). max_mass is the mean over rows and samples of the largest entry in each adjacency row; eff_k is $\exp(\text{row-entropy})$ and measures effective support size.

STAEformer is excluded from the Pearson because its spatial mixing is head-stratified multi-head attention rather than a single row-stochastic adjacency: a max_mass extracted from any one head is not commensurable

with the per-row probability used in the other four backbones, so the proxy-isomorphism assumption underlying the correlation does not hold. The four-backbone Pearson is therefore reported on the row-stochastic subset where the proxy is identically defined.

Under the conservative ranking that treats MegaCRN's negative delta as an adjacency-smoothing artifact (Section 5.3), max_mass and Δ_{sensor} are monotonic across all four models.

F.2.1 Seed-level variation and rank-correlation diagnostics

Per-seed max_mass values for the four adaptive backbones are AGCRN $\{0.0077, 0.0076, 0.0100\}$ (std 0.0014), MegaCRN $\{0.0064, 0.0093, 0.0097\}$ (std 0.0018), MTGNN-Light $\{0.1586, 0.1607, 0.1596\}$ (std 0.0011), and GWNet $\{0.1638, 0.2773, 0.2475\}$ (std 0.0588). Three of the four backbones have per-seed std below the smallest non-tied inter-family gap of 0.07 (MTGNN 0.16 vs. GWNet 0.23); GWNet's larger seed spread is the dominant source of within-family variability.

Per-seed Pearson against Δ_{sensor} is $\{+0.92, +0.99, +0.98\}$ across the three seeds (mean $+0.96$, std 0.04), suggesting that the max_mass relation is not an artifact of seed averaging despite GWNet's larger within-seed spread. Spearman over the four seed-mean values is $\rho_S{=}+0.80$; the single rank exchange swaps AGCRN (0.0085) and MegaCRN (0.0085), whose max_mass values differ by less than 10^{-3}, well below either backbone's per-seed std. Treating the two memory-meta backbones as tied on the proxy yields rank correlation $\rho_S{=}+1.0$ across the three distinguishable groups $\{$tied AGCRN/MegaCRN, MTGNN-Light, GWNet$\}$.

F.3 PEMS04-native routing-concentration replication

To address whether the routing-concentration relation generalizes beyond a single source city, we re-train the four adaptive-graph backbones on PEMS04 (307 sensors, chronological 70/10/20 split) using the same training configuration as the canonical METR-LA bank (Section 3.2). max_mass is then recomputed from each PEMS04-trained checkpoint via the same per-architecture adjacency proxy (Section 5.2), and Δ_{sensor} is measured on PEMS04 test data (3 model seeds $\times$ 5 mask seeds, 30% zero-mask, MAE on kept sensors only).

Two findings strengthen the shared-mechanism hypothesis. (i) max_mass values are tightly preserved across source cities (within a few percent for each backbone), confirming that adjacency concentration is an architecture-level property rather than a dataset artifact. (ii) MegaCRN's METR-LA-specific negative Δ_{sensor} (-0.005) becomes positive on PEMS04 ($+0.057$), consistent with the adjacency-mediated weekday-prior smoothing diagnosis in Section 5.3—the smoothing prior exists in METR-LA but not in the PEMS04 weekday distribution, while the routing-concentration mechanism is shared.

F.4 Theory sketch: first-order exposure bound

Setup. Let a single message-passing layer produce node representations $h_i = \sum_j A_{ij} W x_j$ where $A \in \mathbb{R}^{N\times N}$ is row-stochastic and W is a linear map. Perturbing the inputs at nodes in $\mathcal{I} \subseteq [N]$ by zero-masking, the change in h_i is $\Delta h_i = -\sum_{j\in\mathcal{I}} A_{ij} W x_j$.

Exposure bound. Taking norms and bounding input magnitudes by $\|x\|_\infty$:

$$\|\Delta h_i\|_2 \le \|W\|_2 \cdot \|x\|_\infty \cdot \sum_{j\in\mathcal{I}} A_{ij}.$$

For any row i, let $m_i = \max_j A_{ij}$. Then $\sum_{j\in\mathcal{I}} A_{ij} \le |\mathcal{I}| m_i$, so

$$\|\Delta h_i\|_2 \le \|W\|_2 \cdot \|x\|_\infty \cdot |\mathcal{I}| m_i.$$

Averaging over target nodes yields

$$\frac{1}{N}\sum_i \|\Delta h_i\|_2 \le \|W\|_2 \cdot \|x\|_\infty \cdot |\mathcal{I}| \cdot \text{max_mass}(A).$$

Interpretation. The bound says that concentrated rows have larger worst-case exposure when the dropped set includes high-mass neighbors. It is deliberately local: it does not prove full-horizon MAE behavior and does not cover nonlinear recurrence, temporal convolutions, or decoder amplification. It also should not be read as an exact expected-error result under iid dropout. If each sensor is dropped independently with probability p, then for a row-stochastic A,

$$\mathbb{E}_{\mathcal{I}}\left[\sum_{j\in\mathcal{I}} A_{ij}\right] = p\sum_j A_{ij} = p,$$

which is independent of m_i. Thus max_mass(A) is best viewed as a tail-risk or worst-case exposure proxy. This is the role it plays in Table 25: the empirical ρ=+0.979 on METR-LA and ρ=+0.910 on PEMS04 show that the proxy tracks sensor-dropout fragility across adaptive backbones. The mask-aware intervention (Section 5) targets the same exposure term by zeroing and renormalizing routes attached to known dropped sensors.

G Model Ablations

G.1 GWNet capacity variants (P17 and P21)

We train GWNet[Strong] with hidden dim 128, adaptive embed dim 20, and early-stopping patience 30 (otherwise identical to the standard config). Table 27 compares it to the default GWNet and DCRNN across all shift axes. We additionally report GWNet[Best] (P21), the best configuration identified by in-distribution MAE across a 12-config Stage-1 grid (hidden $\in \{128, 256\}$, embed $\in \{20, 32\}$, lr $\in \{$1e-3, 5e-4, 2e-4$\}$, dropout $\in \{0.1, 0.3\}$); the winning setting is hidden 256, embed 32, lr 5e-4, patience 30.

Doubling hidden dimension and adaptive embedding closes 34% of the in-distribution MAE gap to DCRNN ($7.85 \rightarrow 7.42$ mph) and halves the sensor-dropout Δ ($+0.088 \rightarrow +0.050$), matching the sparse MTGNNLight baseline. However, the cross-city MAE is effectively unchanged ($4.42 \rightarrow 4.46$ mph), the source-only adjacency gap to DCRNN persists (0.262 vs 0.180), and the degree-stratified hub-fragility signature remains monotonic ($1.72\times$ Q4/Q1, still far from DCRNN's $0.95\times$).

The full 12-config sweep GWNet[Best] (hidden 256, embed 32, lr 5e-4, patience 30) further reduces the in-distribution gap (7.22 ± 0.04 mph, closing 63% of the gap to DCRNN) and the sensor-dropout fragility ($+0.042 \pm 0.002$), yet Δ_{sensor} remains $2.1\times$ DCRNN's $+0.020$—confirming P21. The source-only city-transfer normalized MAE (0.262 ± 0.009) is identical to GWNet[Strong]'s (0.262) and still 46% above DCRNN's 0.180. Capacity attenuates but does not eliminate the architectural fragility, supporting the mechanism interpretation in Section 5.

G.2 MegaCRN memory-bank ablation

To test whether MegaCRN's meta-memory bank absorbs missing sensors as a form of implicit regularization, we zero-mask and re-route its attention/memory paths at inference in four modes. *full*: the unmodified model; *memzero*: query the memory bank with a zero vector so the prototype fall-back is fixed regardless of input; *graphonly*: zero out the attention output keeping the adjacency pathway; *attnonly*: use identity adjacency while keeping memory attention. We report Δ_{sensor} means over three seeds and five dropout ratios $r \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$.

memzero deepens rather than flattens the U-shape, refuting the memory-bank regularization hypothesis. **attnonly** (identity adjacency) eliminates Δ_{sensor} completely, establishing that cross-node flow through the adjacency is the sole driver. Per-seed σ at r=0.2 is 0.031, approximately equal to the mean—the apparent robustness is seed-fragile. The mechanism is adjacency-mediated smoothing: zero-diffused predictions drift toward a weekday-mean prior that accidentally tracks the test set.

H Architecture-Selection Decision Tree

Table 29 reports the conservative $2\times2\times2$ recommendation table over the deployment profile. The tree is generated by `experiments/decision_tree_artifact.py` from the same per-model normalized fragility vectors used in the main paper. The conservative ranking demotes flagged *(model, axis)* pairs where low apparent fragility is an artifact (*(megacrn, sensor)*; see Section 5.3).

Four of the eight cells reach DCRNN under multi-shift risk, reflecting DCRNN's uniformly low fragility across axes. MegaCRN is recommended only under the single-shift city-transfer profile; STAEformer wins the sensor-only cell; AGCRN wins the two weekend-exposed cells without city transfer. The tree is released as a pip-installable anonymous artifact (`stgnn-stress`) so that practitioners can reproduce the recommendation and the underlying rankings on new frozen checkpoints.

Table 7: Pre-registered hypotheses and outcomes. Confirmed (✓), partially confirmed, or refuted (×) by experiment.

Prediction	**Outcome**	**Result**
GWNet fails hardest under city transfer (topology overfitting)	GWNet is worst under sensor dropout, not city transfer; GRU is worst at city transfer	×
Graph-based models are more robust than GRU to sensor dropout	GRU is nearly as robust as DCRNN at 30%; adaptive graph hurts, fixed graph helps	Partial ×
All models degrade on PEMS-BAY vs. METR-LA	Source-stat MAE improves on PEMS-BAY (simpler traffic); target-stat reveals real gaps	×
Adaptive adjacency drives sensor-dropout fragility ($\geq$30% reduction when ablated)	43.1% reduction confirmed	✓
Temporal shift magnitude differs across architectures	All architectures within one std of each other	×
STGCN (fixed spectral adj) $\Delta_{\text{sensor}} \leq$ DCRNN (fixed diffusion adj)	STGCN $+0.078 >$ DCRNN $+0.020$; but both $<$ GWNet $+0.088$. Adaptive adj is the dominant driver.	×
P11–P13 added prior to structured / coverage experiments:		
[P11] GWNet transfer gap persists (GWNet worst) across 150- and 207-node PEMS-BAY coverage	Confirmed: ranking preserved; GWNet +41% MAE at 150 nodes vs. -2% for DCRNN	✓
[P12] Hub dropout raises GWNet Δ_{sensor} relative to uniform random (dense routing)	Confirmed: GWNet $+38\%$ ($+0.127$); DCRNN lower ($+0.008$ vs. random $+0.021$)	✓
[P13] Mask-aware recovery meaningful with FP $\leq 5\%$, FN $\leq 10\%$ detection	Partial: 27% recovery at FP=5%/FN=10%; high variance at FP=10%/FN=20%	Partial ✓
P14–P16 added prior to inductive-baseline and capacity-matched experiments:		
[P14] MTGNNLight (sparse cosine-sim) shows lower Δ_{sensor} than GWNet (dense product)	Confirmed: MTGNNLight $+0.051$, GWNet $+0.088$ (42% lower)	✓
[P15] GWNet[CM] (doubled patience) shows same Δ_{sensor} as standard GWNet	Confirmed: $+0.088$ vs $+0.088$ (identical); fragility is architectural not training artifact	✓
[P16] Source-only adj control narrows but does not eliminate DCRNN's transfer advantage	Confirmed: DCRNN 0.180 vs GWNet 0.268 (49% gap remains under fair comparison)	✓
P17–P19 added prior to stronger-GWNet, second-city-pair, and per-horizon experiments:		
[P17] GWNet[Strong] (hidden 128, embed 20) still shows $\Delta_{\text{sensor}}>2\times$ DCRNN and unchanged city MAE	Confirmed: $+0.050$ vs $+0.020$ ($2.5\times$); city 4.46 vs 4.42 mph	✓
[P18] DCRNN is best-generalizing on ≥ 1 PEMS04-source direction (PEMS04$\rightarrow$METR-LA, PEMS04$\rightarrow$PEMS-BAY)	Confirmed on *both*: DCRNN 7.01/2.65 vs next-best GRU 11.03/GWNet 4.58	✓
[P19] GWNet's per-horizon city-transfer MAE is flatter than DCRNN's (static mis-routing, not accumulation)	Confirmed: GWNet $+4\%$ h1$\rightarrow$h12 growth vs DCRNN $+46\%$	✓
P21 added prior to full-sweep best-config experiment:		
[P21] GWNet[Best] (12-config sweep: hidden 256, embed 32, lr 5e-4) still shows $\Delta_{\text{sensor}}>2\times$ DCRNN	Confirmed: $+0.042$ vs $+0.020$ ($2.1\times$); in-dist 7.22 mph; optimal training cannot eliminate routing fragility	✓

Table 8: Estimand-level crosswalk: which numbers are cross-family-comparable.

Estimand	Source	Unit	Normalization	Cross-family?
Clean MAE	T.2, T.4	mph (canon.) / masked (ext.)	per-family	No
Δ_{sensor}	T.2, T.4	normalized	source-stat	Yes (within-family)
Cross-city MAE	T.2, T.4	normalized	source / target (both)	Yes (matched-access)
Δ_{temp}	T.2, T.4	normalized	source-stat	Yes
Routing concentration	T.25, App. F.3	unitless $[0, 1]$	per-row	Yes (proxy-matched, n=4)

Table 9: **GWNet is the most fragile backbone at every dropout level; the gap grows nonlinearly.** Δ_{sensor} mean over 3 seeds across dropout ratios 10–50%. GWNet/DCRNN ratio (bottom) quantifies relative fragility.

Backbone	10%	20%	30%	40%	50%
DCRNN	+0.005	+0.012	+0.020	+0.033	+0.044
GWNet	+0.013	+0.044	+0.088	+0.149	+0.189
GRU	+0.001	+0.005	+0.025	+0.073	+0.088
GWNet/DCRNN	2.5×	3.8×	4.3×	4.6×	4.3×

Table 10: Multi-mask sensor dropout: Δ_{sensor} mean and std over 5 random masks (seeds 42–46), then averaged over 3 model seeds. Most values are consistent with primary-mask results in Table 2; GRU is the exception noted in the protocol.

Backbone	Δ_{sensor} (multi-mask mean)	Seed std	vs. Table 2
DCRNN (fixed diffusion)	+0.021	±0.002	+0.020 (consistent)
GRU (graph-free)	+0.044	±0.002	+0.025 (seed-42 was low-variance mask)
MTGNNLight (sparse cosine-sim)	+0.051	±0.012	+0.051 (consistent)
STGCN (fixed spectral)	+0.079	±0.006	+0.078 (consistent)
GWNet (adaptive adj)	+0.092	±0.030	+0.088 (consistent; high mask variance)
GWNet[CM] (patience=30)	+0.088	±0.017	+0.088 (P15 confirmed identical)

Table 11: Structured vs. uniform sensor dropout (Δ_{sensor}, mean over 3 seeds, 30% drop rate). Hub: top-k degree sensors dropped. Contiguous: BFS cluster from highest-degree seed. GWNet ordering is preserved (>STGCN>DCRNN); structured patterns amplify GWNet's fragility but reduce DCRNN/STGCN's apparent sensitivity, because hub dropout keeps peripheral (easier-to-predict) sensors.

Backbone	Random (uniform)	Hub dropout	Contiguous dropout
DCRNN (fixed diffusion)	+0.021	+0.008	+0.005
STGCN (fixed spectral)	+0.079	+0.026	+0.022
GWNet (adaptive adj)	+0.088	+0.127	+0.096

Table 12: Sensor-dropout Δ_{sensor} by node-degree quartile (30% zero-mask, kept sensors only, 3-seed mean). Mean node degrees: Q1 $\bar{d} = 1.83$, Q2 $\bar{d} = 3.32$, Q3 $\bar{d} = 4.48$, Q4 $\bar{d} = 5.98$.

Backbone	Q1 (low)	Q2	Q3	Q4 (high)	Q4/Q1
DCRNN	+0.019	+0.026	+0.018	+0.018	0.95×
MTGNNLight	+0.044	+0.052	+0.050	+0.055	1.25×
STGCN	+0.059	+0.089	+0.074	+0.089	1.50×
GWNet	+0.066	+0.076	+0.085	+0.121	1.83×

Table 13: Mask-aware GWNet sensitivity to imperfect mask detection (3 model seeds). FP = fraction of healthy sensors erroneously zeroed in the adjacency; FN = fraction of failed sensors left un-zeroed. Δ_{sensor}: estimated as $\Delta^{\text{perfect}}_{\text{sensor}} + \delta_{\text{from perfect}}$ where $\Delta^{\text{perfect}}_{\text{sensor}} = +0.023$ (from Table 6) and $\delta_{\text{from perfect}}$ is the mean degradation relative to perfect detection.

Detector scenario	**FP / FN**	$\delta_{\text{from perfect}}$ **(mean)**	**Recovery vs. no mask-aware**
Standard (no mask-aware)	—	—	0%
Perfect mask	0% / 0%	0.000	75%
Conservative detector	5% / 10%	+0.044	∼27%
Noisy detector	10% / 20%	$+0.077^{*}$	∼17%

Table 14: Mask-aware GWNet recovery is stable across dropout ratios and structured patterns. "Standard" = silent zero-mask; "mask-aware" = adjacency rows/cols zeroed and renormalized for dropped sensors; recovery $= 1 - \Delta^{\text{mask-aware}}_{\text{sensor}} / \Delta^{\text{standard}}_{\text{sensor}}$. Script: `experiments/mask_aware_dropout_sweep.py`; data: `results/polish_v15/mask_aware_ratio_sweep.json`.

Cell	$\Delta^{\text{std.}}_{\text{sensor}}$	$\Delta^{\text{mask-aware}}_{\text{sensor}}$	**Recovery**
Random 5%	$+0.004 \pm 0.002$	$+0.000 \pm 0.001$	$109\% \pm 29$
Random 10%	$+0.019 \pm 0.006$	$+0.005 \pm 0.002$	$76\% \pm 6$
Random 20%	$+0.051 \pm 0.013$	$+0.015 \pm 0.002$	$70\% \pm 5$
Random 30% (main)	$+0.092 \pm 0.030$	$+0.023 \pm 0.009$	$75\% \pm 3$
Random 40%	$+0.145 \pm 0.026$	$+0.035 \pm 0.006$	$75\% \pm 3$
Hub-30% (structured)	$+0.116 \pm 0.037$	$+0.034 \pm 0.001$	71%

Table 15: Sensor-dropout Δ under three imputation strategies (30% drop rate, seed=42, 3 model seeds). LOCF improvement: fraction of zero-mask gap closed by LOCF. Negative improvement indicates worse than zero-mask.

Backbone	Δ_{zero}	Δ_{LOCF}	Δ_{mean}	**LOCF improvement**
DCRNN	+0.020	≈ 0	+0.025	$\sim 102\%$
MTGNNLight	+0.051	≈ 0	+0.077	$\sim 100\%$
STGCN	+0.078	+0.003	+0.079	$\sim 97\%$
GWNet	+0.088	≈ 0	+0.109	$\sim 101\%$

Table 16: Cross-city MAE under two normalization conventions. Source-stat: METR-LA train mean/std for both normalization and denormalization (lower = PEMS-BAY traffic is statistically simpler in normalized space). Target-stat: PEMS-BAY train statistics for denormalization (interpretable mph units). Mean $\pm$ population std over 5 random 207-node subsets and 3 model seeds.

Backbone	**Clean MAE (METR-LA, norm)**	**Source-stat MAE (norm)**	**Target-stat MAE (mph)**
DCRNN (fixed diffusion)	0.359	0.168	2.74 ± 0.01
STGCN (fixed spectral)	0.388	0.209	3.27 ± 0.02
GWNet (adaptive adj)	0.427	0.268	4.42 ± 0.15
GRU (graph-free)	0.408	0.278	4.73 ± 0.19

Table 17: Cross-city transfer MAE (target-stat, mph) under variable sensor coverage. 150-node: 150 PEMS-BAY sensors active, 57 input slots zero-padded to 207. 207-node: full model capacity (different subset draws from main table; see text). Mean $\pm$ std over 3 random PEMS-BAY subsets, 3 model seeds.

Backbone	**150-node coverage**	**207-node (full)**
DCRNN (fixed diffusion)	4.25 ± 0.03	4.33 ± 0.05
STGCN (fixed spectral)	4.94 ± 0.25	5.07 ± 0.33
GWNet (adaptive adj)	7.49 ± 1.08	5.30 ± 0.25

Table 18: Full cross-city transfer comparison. *Source-only (norm)*: all models constrained to METR-LA adjacency; canonical main-table values. *Target-adj (norm)*: DCRNN/STGCN use PEMS-BAY adjacency (GWNet/MTGNNLight/GRU unchanged). *Target-adj (mph)*: same target-adj evaluation, denormalized using PEMS-BAY statistics. Mean ± population std over 5 PEMS-BAY subsets, 3 model seeds.

Backbone	**Source-only (norm) [main table]**	**Target-adj (norm)**	**Target-adj (mph)**
DCRNN (fixed diffusion)	0.180 ± 0.001	0.167 ± 0.002	2.74 ± 0.01
MTGNNLight (cosine-sim)	0.244 ± 0.004	0.244 ± 0.004	4.15 ± 0.11
STGCN (fixed spectral)	0.249 ± 0.008	0.202 ± 0.005	3.27 ± 0.02
GRU (graph-free)	0.278 ± 0.009	0.278 ± 0.009	4.73 ± 0.19
GWNet (dense adaptive)	0.268 ± 0.006	0.268 ± 0.006	4.42 ± 0.15

Table 19: PEMS04-source cross-city transfer (3 seeds × 5 target subsets, mean ± std). PEMS04 in-dist MAE in native flow units; target MAE in target-stat mph. DCRNN is best in *both* new directions by $> 37\%$ margin.

Backbone	**PEMS04 in-dist**	**→ METR-LA (mph)**	**→ PEMS-BAY (mph)**
DCRNN	26.19 ± 0.02	$\mathbf{7.01 \pm 0.09}$	$\mathbf{2.65 \pm 0.02}$
GWNet	33.47 ± 0.45	11.39 ± 0.62	4.58 ± 0.16
GWNet[Strong]	32.62 ± 0.17	12.58 ± 0.32	4.61 ± 0.02
GRU	25.81 ± 0.06	11.03 ± 0.01	4.68 ± 0.03

Table 20: Cross-city MAE (mph) on METR-LA target with the existing index-repetition padding vs. a no-pad variant that scores only the 207 unique target nodes. PEMS04 → PEMS-BAY (N=325) is unaffected since target > source (subsample only). 3 PEMS04 seeds × 5 target subsets. Script: `experiments/cross_city_no_pad_sensitivity.py`; data: `results/polish_v15/cross_city_no_pad_sensitivity.json`.

Backbone	**Padded MAE (mph)**	**No-pad MAE (mph)**	**Δ (pad − nopad)**
DCRNN	7.014 ± 0.085	7.166 ± 0.088	-0.152
STGCN	7.176 ± 0.169	7.295 ± 0.158	-0.118
GWNet	11.390 ± 0.623	12.051 ± 0.422	-0.661
MTGNN	10.292 ± 0.090	11.020 ± 0.083	-0.728
GRU	11.026 ± 0.008	11.756 ± 0.012	-0.730

Table 21: Per-horizon normalized MAE on METR-LA at h=1, 6, 12. Clean and 30% sensor dropout computed on the METR-LA test set; city-src computed on the 5-subset PEMS-BAY evaluation. Δh: relative growth from h=1 to h=12.

	Clean			**Masked 30%**			**City-src**		
Backbone	h_1	h_6	h_{12}	h_1	h_6	h_{12}	h_1	h_6	h_{12}
DCRNN	0.251	0.356	0.448	0.356	0.449	0.527	0.133	0.160	0.194
MTGNNLight	0.291	0.370	0.450	0.371	0.444	0.515	0.237	0.240	0.253
STGCN	0.296	0.385	0.466	0.438	0.513	0.578	0.173	0.199	0.230
GRU	0.331	0.403	0.474	0.375	0.435	0.499	0.263	0.269	0.282
GWNet	0.363	0.423	0.485	0.464	0.516	0.568	0.261	0.264	0.272
GWNet[Strong]	0.333	0.401	0.468	0.390	0.457	0.522	0.254	0.257	0.260

Table 22: Weekday → weekend Δ_{temp} (normalized) under covariate interventions. Script: `experiments/temporal_covariate_ablation.py`.

Model	**Clean**	**Randomize**	**Swap**	**Reduction (swap)**
MegaCRN	0.7336	0.7254	0.7336	0.0%
STAEformer	0.7495	0.7212	0.6556	12.5%

Table 23: Density sweep anchor points: relative fragility increases monotonically with adaptive connectivity density. Rel. fragility $= \Delta_{\text{sensor}}/\text{MAE}_{\text{in-dist}}$. Mean $\pm$ std over 3 seeds; Δ_{sensor} averaged over 5 random 30% masks.

Variant	**In-dist MAE**	Δ_{sensor} **(30%)**	**Rel. fragility**
GWNet[fixed]	8.27 ± 0.08	$+0.050 \pm 0.010$	0.60%
GWNet[k=10]	9.03 ± 0.42	$+0.073 \pm 0.030$	0.80%
GWNet[k=20]	8.85 ± 0.34	$+0.074 \pm 0.032$	0.84%
GWNet[k=full]	7.85 ± 0.11	$+0.092 \pm 0.030$	1.17%
MTGNNLight (ref)	6.88 ± 0.11	$+0.051 \pm 0.012$	0.74%

Table 24: Full k-grid density sweep for GWNet[adaptive] on METR-LA. Mean $\pm$ std over 3 seeds. Δ_{sensor}: averaged over 5 random 30% zero-masks (seeds 42–46). Rel. fragility $= \Delta_{\text{sensor}}/\text{MAE}_{\text{in-dist}}$. Note: $k<$full disrupts embeddings trained for dense routing, causing in-distribution degradation at low k; the clean-performance cost at k=5 is $+17\%$ relative to k=full.

k	**In-dist MAE**	Δ_{sensor} **(30%)**	**Rel. fragility**
5	9.21 ± 0.50	$+0.072 \pm 0.031$	0.78%
10	9.03 ± 0.42	$+0.073 \pm 0.030$	0.80%
15	8.94 ± 0.38	$+0.073 \pm 0.031$	0.81%
20	8.85 ± 0.34	$+0.074 \pm 0.032$	0.84%
30	8.66 ± 0.30	$+0.078 \pm 0.033$	0.89%
full $= 207$	7.85 ± 0.11	$+0.092 \pm 0.030$	1.17%

Table 25: Routing-sparsity spectrum vs. sensor-dropout fragility. Pearson correlation of max_mass and Δ_{sensor} across four adaptive models: $\rho = +0.979$. Script: `experiments/routing_sparsity_spectrum.py`. Δ_{sensor} values are the routing-spectrum estimates used for this correlation.

Model	**Family**	**max_mass**	**eff**_k	**density**	Δ_{sensor}
AGCRN	dense-symmetric	0.008	≈ 200	1.00	$+0.011$
MegaCRN	memory-meta	0.008	≈ 200	1.00	-0.005
MTGNN-Light	sparse-top-k	0.160	—	sparse	$+0.051$
GWNet	dense-bilinear	0.230	≈ 114	1.00	$+0.092$

Table 26: Routing-concentration relation replicates on PEMS04. Pearson $\rho(\text{max_mass}, \Delta_{\text{sensor}}) = +0.910$ across the four adaptive backbones (METR-LA reference: $+0.979$). Native checkpoints; same protocol as Table 25. Script: `experiments/pems04_routing_gate_native.py`; data: `results/polish_v15/pems04_native_routing_gate.json`.

	METR-LA (reference)		**PEMS04 (native)**	
Model	**max_mass**	Δ_{sensor}	**max_mass**	Δ_{sensor}
AGCRN	0.008	$+0.011$	0.006 ± 0.000	$+0.051 \pm 0.006$
MegaCRN	0.008	-0.005	0.005 ± 0.000	$+0.057 \pm 0.018$
MTGNN	0.160	$+0.051$	0.145 ± 0.001	$+0.119 \pm 0.004$
GWNet	0.230	$+0.092$	0.248 ± 0.020	$+0.114 \pm 0.025$

Table 27: GWNet capacity scaling: standard, P17 strong, and P21 best-sweep configs vs. DCRNN (3 seeds, mean $\pm$ std). Source-only adjacency MAE follows the P16 fairness control protocol. Q4/Q1 is the degree-stratified fragility ratio (Appendix C.4). GWNet[Best] city MAE (mph) and Q4/Q1 were not separately evaluated in the sweep; source-only normalized values are reported.

Variant	**In-dist (mph)**	Δ_{sensor} **(30%)**	**City MAE (mph)**	**Source-only adj (norm)**	**Q4/Q1 ratio**
DCRNN	6.58 ± 0.03	$+0.020 \pm 0.002$	2.74 ± 0.01	0.180 ± 0.001	$0.95\times$
GWNet	7.85 ± 0.11	$+0.088 \pm 0.017$	4.42 ± 0.15	0.268 ± 0.006	$1.83\times$
GWNet[Strong]	7.42 ± 0.04	$+0.050 \pm 0.018$	4.46 ± 0.06	0.262 ± 0.004	$1.72\times$
GWNet[Best]	7.22 ± 0.04	$+0.042 \pm 0.002$	—	0.262 ± 0.009	—

Table 28: MegaCRN Δ_{sensor} (normalized) by dropout ratio and ablation mode. Script: `experiments/megacrn_memory_ablation.py`.

Mode	r=0.1	r=0.2	r=0.3	r=0.4	r=0.5
full	-0.001	-0.043	-0.040	-0.016	$+0.045$
memzero	-0.006	-0.067	-0.075	-0.038	-0.007
graphonly	$+0.001$	-0.027	-0.024	$+0.012$	$+0.068$
attnonly	0.000	0.000	0.000	0.000	0.000

Table 29: Architecture recommendation per deployment profile. Margin is the difference in weighted-risk score to the runner-up (lower = closer call). Confidence is margin normalized by runner-up risk.

sensor	**city**	**weekend**	**Recommended**	**Runner-up**	**Confidence**
no	no	no	DCRNN	AGCRN	0.30
no	no	yes	AGCRN	MTGNN-Light	1.00
no	yes	no	MegaCRN	DCRNN	1.00
no	yes	yes	DCRNN	AGCRN	0.51
yes	no	no	STAEformer	AGCRN	0.18
yes	no	yes	AGCRN	DCRNN	0.19
yes	yes	no	DCRNN	MegaCRN	0.34
yes	yes	yes	DCRNN	AGCRN	0.30

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes].

   Justification: The abstract and introduction state that the paper is a pre-registered diagnostic of frozen traffic STGNNs, not a new model or dataset. The scope is limited to the evaluated shifts, source cities, and masking assumptions.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes].

   Justification: Section 6 states the limits of the model set, single-source training, silent zero-masking assumption, and unresolved MegaCRN temporal mechanism; Appendix B.2 summarizes reproducibility and asset scope.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes].

   Justification: The only theoretical component is a first-order exposure bound. Section 5 states its local interpretation, and Appendix F.4 gives the assumptions, derivation, and iid-dropout caveat.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes].

   Justification: Sections 3–5 describe the datasets, splits, frozen-checkpoint protocol, shifts, metrics, and interventions. Appendix B.2 identifies the canonical result files, registry, configs, and figure-generation commands.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes].

   Justification: The submission includes evaluation scripts, configs, result tables, and the recommendation artifact. The raw traffic datasets are public third-party datasets and are cited rather than redistributed.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes].

   Justification: Section 3 reports the model bank, hyperparameters, seeds, splits, normalization, and all three shift protocols; appendix tables give additional protocol variants.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes].

   Justification: Main tables and figures report mean $\pm$ population standard deviation over three seeds unless otherwise stated; appendix tables specify when additional mask seeds or target subsets are included.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes].

Justification: Appendix B.2 states that runs use one CUDA GPU per job and reports representative logged evaluation times. The paper also gives the frozen-checkpoint protocol so reviewers can verify results without retraining all models.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes].

   Justification: The work uses public traffic benchmark data, reports negative and refuted results, and does not involve human subjects, private personal data collection, or deployment decisions.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes].

    Justification: Section 6 discusses deployment guidance and limitations. The main positive impact is safer model selection under traffic-sensor failures; the main risk is overgeneralizing the recommendations beyond the evaluated cities and masking model, which the paper explicitly limits.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A].

    Justification: The paper does not release high-risk generative models, scraped datasets, or sensitive personal data. It uses public traffic forecasting benchmarks and lightweight evaluation artifacts.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes].

    Justification: The paper cites the source papers for METR-LA, PEMS-BAY, PEMS04, and all model families. Appendix B.2 states that raw datasets are not redistributed and remain under their original terms.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes].

    Justification: The new assets are evaluation scripts, canonical result tables, and a small recommendation artifact rather than a new dataset. Appendix B.2 documents their role and file locations.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A].

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    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

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